

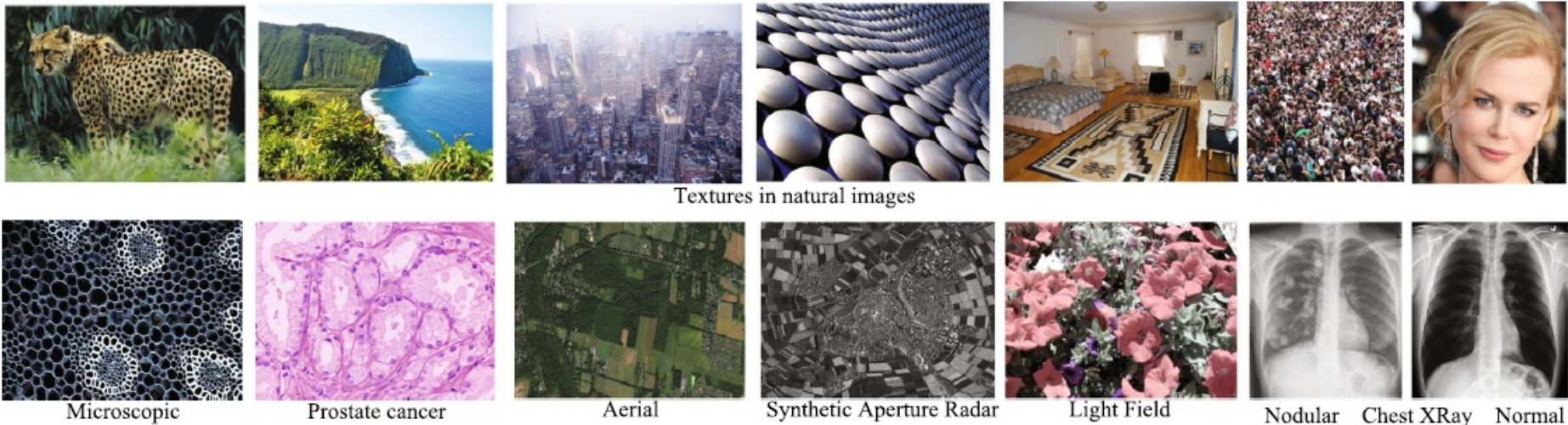
COMP 3411/9414

Artificial Intelligence

AI in Computer Vision – Part 2

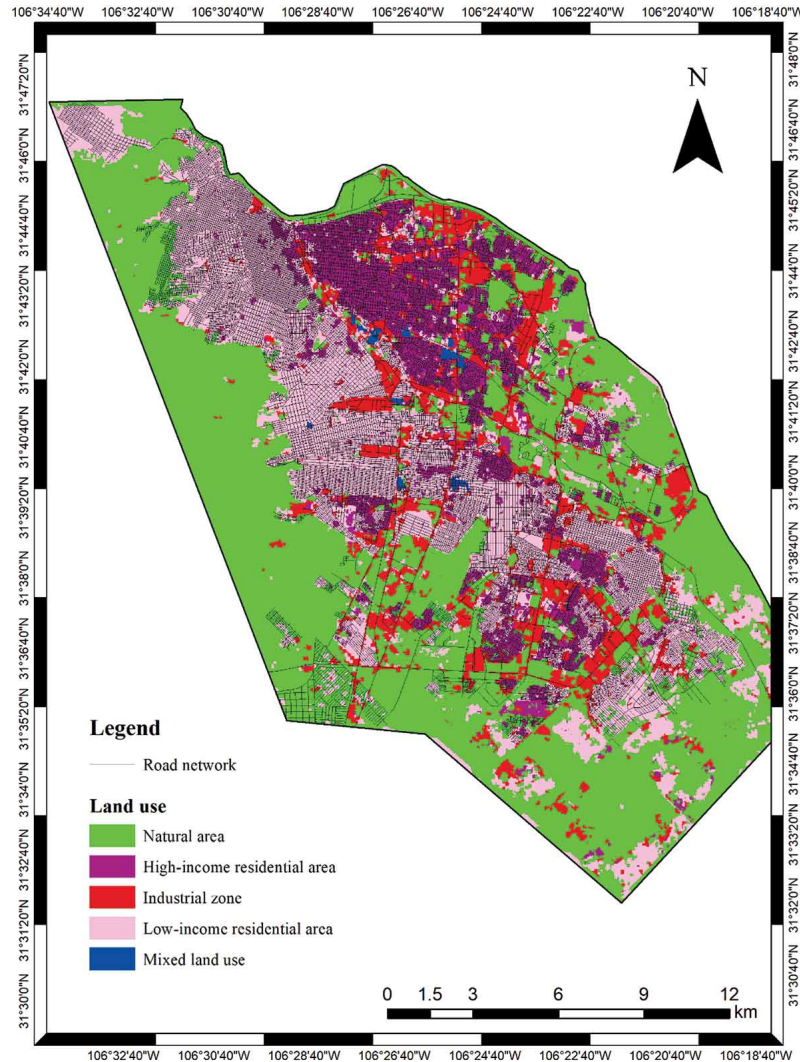
Texture Classification

- Objective of study:
 - Classifying/recognising different texture patterns automatically
 - It's been a fundamental computer vision problem with active research and industry development for multiple decades.
 - Real life applications in many different fields



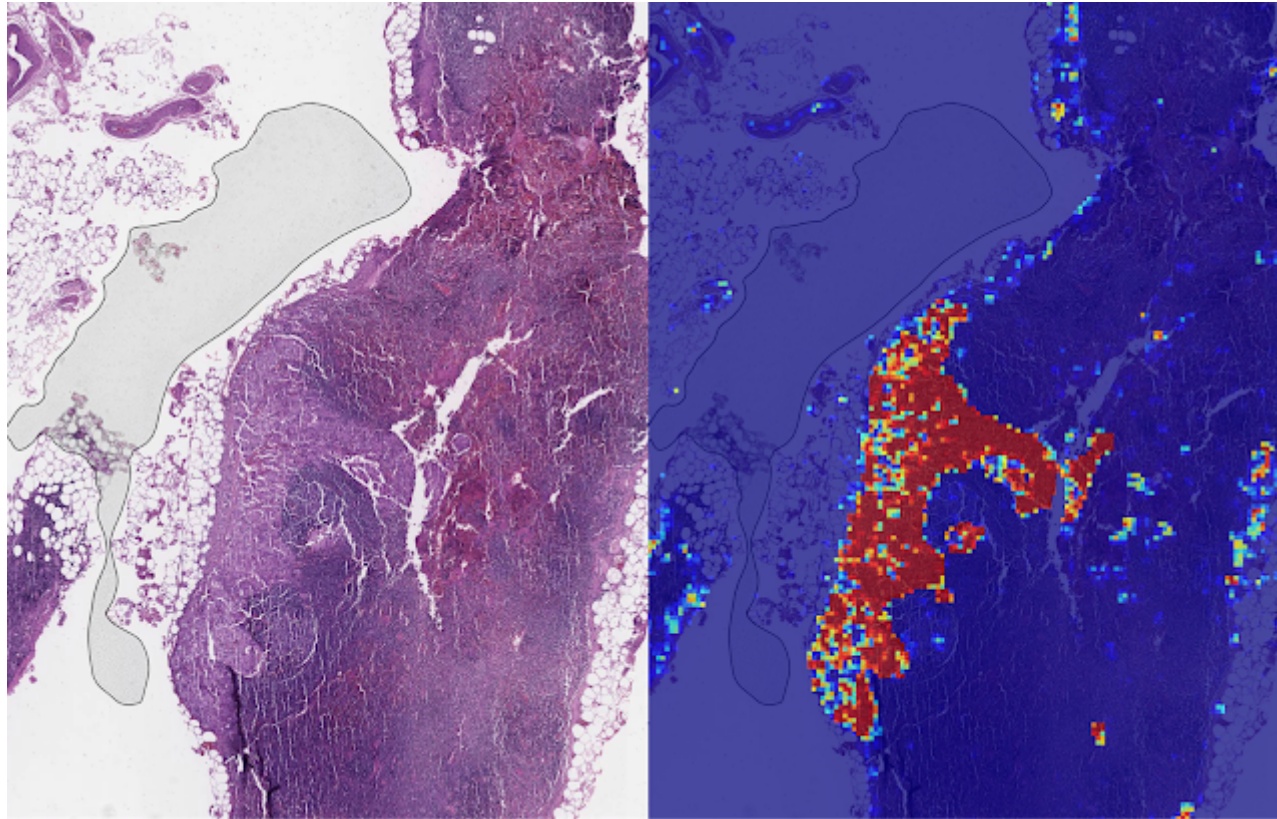
Source: L. Liu et al. *From BoW to CNN: two decades of texture representation for texture classification*.
International Journal of Computer Vision, 2019.

Example – urban planning



Source: I. Hernandez & W. Shi. *A random forests classification method for urban land-use mapping integrating spatial metrics and texture analysis*. International Journal of Remote Sensing, 2018.

Example – cancer detection



Source: <https://ai.googleblog.com/2018/10/applying-deep-learning-to-metastatic.html>

Example – cancer detection



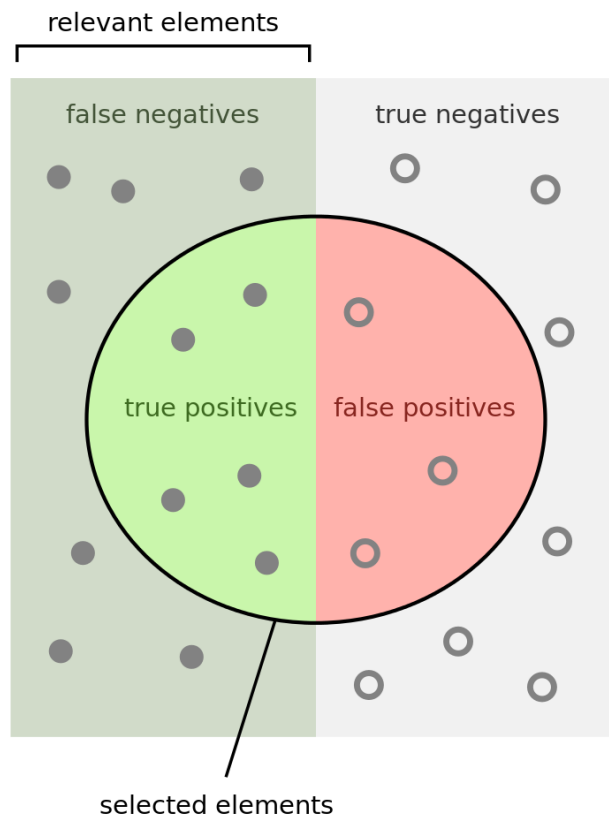
Source: <https://ai.googleblog.com/2018/10/applying-deep-learning-to-metastatic.html>

Texture Classification

- How to solve this problem with artificial intelligence?
 - Pre-processing?
 - Feature representation?
 - Classifier?
 - Training data?
 - Evaluation metrics?

Evaluation Metrics

- Classification is often evaluated using *Precision* and *Recall*.



How many selected items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

Source: https://en.wikipedia.org/wiki/Precision_and_recall

Classifier

- K-nearest neighbour
- Linear regression
- Decision trees
- Artificial Neural Networks
- Support vector machine
- ...
- A good introduction:
<https://www.dezyre.com/article/top-10-machine-learning-algorithms/202>

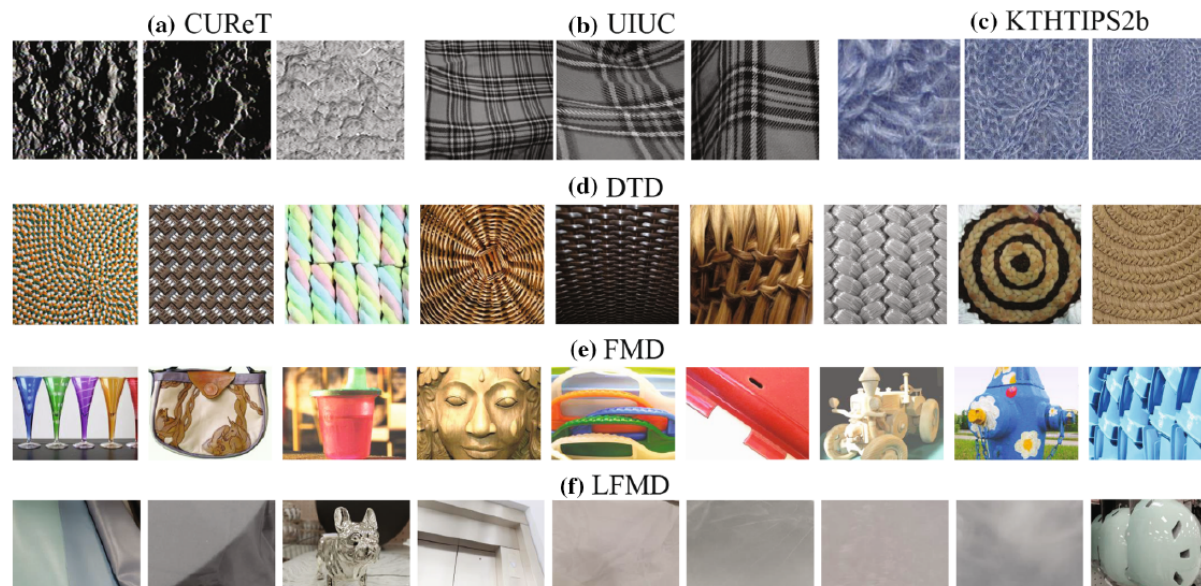
Training Data

- Six well-known public datasets are commonly used for benchmarking

Name	# classes	# images
CUReT	92	5612
UIUC	25	1000
KTHTIPS2b	11	4752
FMD	10	1000
LFMD	12	1200
DTD	47	5640

Feature Representation

- How to encode the texture information effectively so that the images can be classified accurately?
- Critical problems:
 - intra-class variation and inter-class ambiguity



Feature Representation

- Feature representation methods evolve over time:
 - Since 1970s: morphological feature descriptors (Haralick, Gabor)
 - Since 2000: high-dimensional texture descriptors (LBP, HOG, SIFT)
 - Also since 2000: feature encoding (BoW, Texton, FV)
 - Since 2012: deep learning (AlexNet, VGG, ResNet)
- All methods are widely used in current studies.
- The latter two are learning-based approaches.
- Still a difficult problem with low classification accuracies.

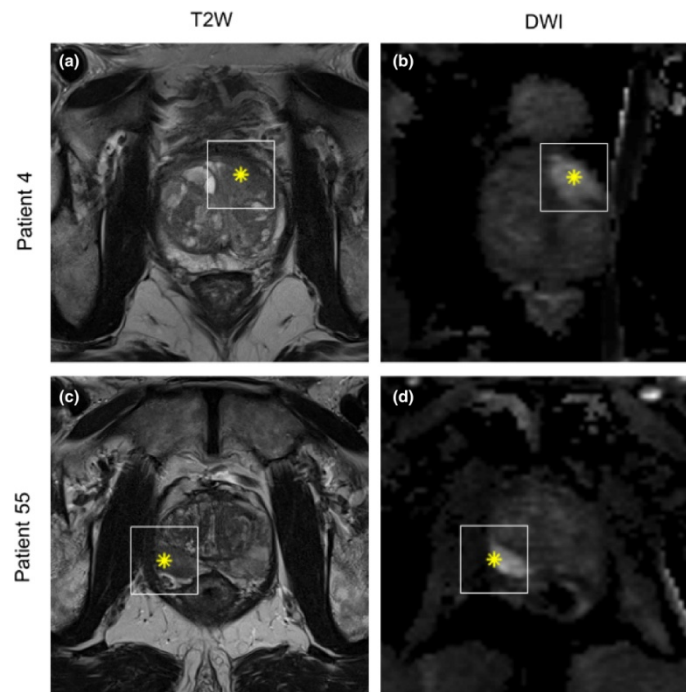
Feature Descriptors

- Haralick features:
 - R. Haralick and I. Dinstein. *Textural features for image classification*. IEEE Trans Syst Man Cybern, 1973.
 - A set of morphological features, to describe the texture features with meaningful concepts (e.g. contrast, variance, entropy)

Angular Second Moment	$\sum_i \sum_j p(i, j)^2$	Sum Variance	$\sum_{i=2}^{2N_g} (i - f_s)^2 p_{x+y}(i)$
Contrast	$\sum_{n=0}^{N_g-1} n^2 \{ \sum_{i=1}^{N_g} \sum_{j=1}^{N_g} p(i, j) \}, i - j = n$	Sum Entropy	$-\sum_{i=2}^{2N_g} p_{x+y}(i) \log\{p_{x+y}(i)\} = f_s$
Correlation	$\frac{\sum_i \sum_j (ij) p(i, j) - \mu_x \mu_y}{\sigma_x \sigma_y}$ where μ_x , μ_y , σ_x , and σ_y are the means and std. deviations of p_x and p_y , the partial probability density functions	Entropy	$-\sum_i \sum_j p(i, j) \log(p(i, j))$
Sum of Squares: Variance	$\sum_i \sum_j (i - \mu)^2 p(i, j)$	Difference Variance	$\sum_{i=0}^{N_g-1} i^2 p_{x-y}(i)$
Inverse Difference Moment	$\sum_i \sum_j \frac{1}{1+(i-j)^2} p(i, j)$	Difference Entropy	$-\sum_{i=0}^{N_g-1} p_{x-y}(i) \log\{p_{x-y}(i)\}$
Sum Average	$\sum_{i=2}^{2N_g} i p_{x+y}(i)$ where x and y are the coordinates (row and column) of an entry in the co-occurrence matrix, and $p_{x+y}(i)$ is the probability of co-occurrence matrix coordinates summing to $x + y$	Info. Measure of Correlation 1	$\frac{HXY - HXY1}{\max\{HX, HY\}}$
		Info. Measure of Correlation 2	$(1 - \exp[-2(HXY2 - HXY)])^{\frac{1}{2}}$ where $HXY = -\sum_i \sum_j p(i, j) \log(p(i, j))$, HX , HY are the entropies of p_x and p_y , $HXY1 =$ $-\sum_i \sum_j p(i, j) \log\{p_x(i) p_y(j)\}$ $HXY2 =$ $-\sum_i \sum_j p_x(i) p_y(j) \log\{p_x(i) p_y(j)\}$
		Max. Correlation Coeff.	Square root of the second largest eigenvalue of $\mathbf{Q}$ where $\mathbf{Q}(i, j) = \sum_k \frac{p(i, k) p(j, k)}{p_x(i) p_y(k)}$

Feature Descriptors

- Haralick features:
 - Commonly used nowadays in medical imaging studies due to its simplicity and interpretability

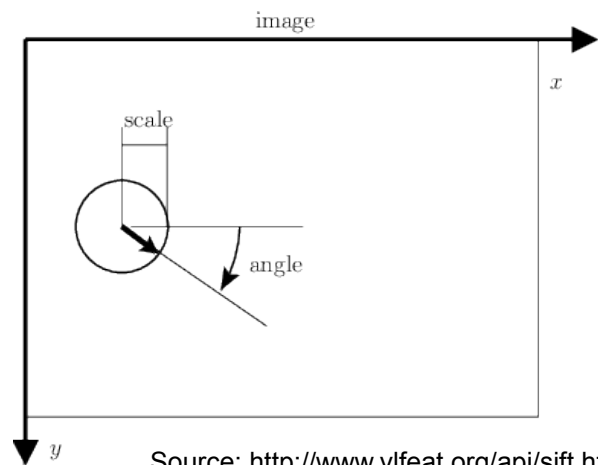


C. Jensen et al. *Assessment of prostate cancer prognostic Gleason grade group using zonal-specific features extracted from biparametric MRI using a KNN classifier*. Journal of Applied Clinical Medical Physics, 2019.

1. Pre-processing
2. Extract Haralick, run-length and histogram features from the region of interest
3. Feature selection
4. Classification using kNN

Feature Descriptors

- Scale-invariant feature transform (SIFT):
 - D. Lowe. *Distinctive image features from scale-invariant keypoints*. International Journal of Computer Vision, 2004.
 - Probably the most popular texture feature descriptor prior to the success of deep learning
 - Moved away from meaningful descriptions, but more numerical, abstract and high-dimensional

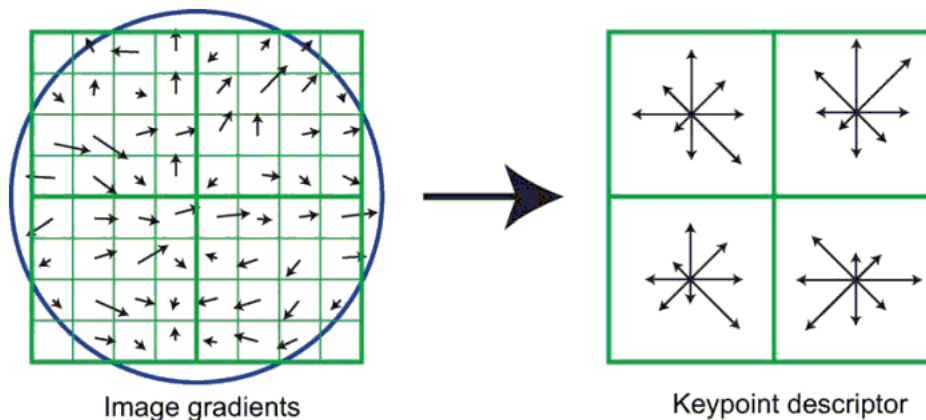


A SIFT keypoint is a circular image region with an orientation.

Source: <http://www.vlfeat.org/api/sift.html>

Feature Descriptors

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A keypoint descriptor is 128-dimensional (i.e. it's a vector of 128 elements).

Feature Descriptors

- Scale-invariant feature transform (SIFT):
 - Often used to perform image matching



Source: <http://www.vlfeat.org/overview/sift.html>

Feature Descriptors

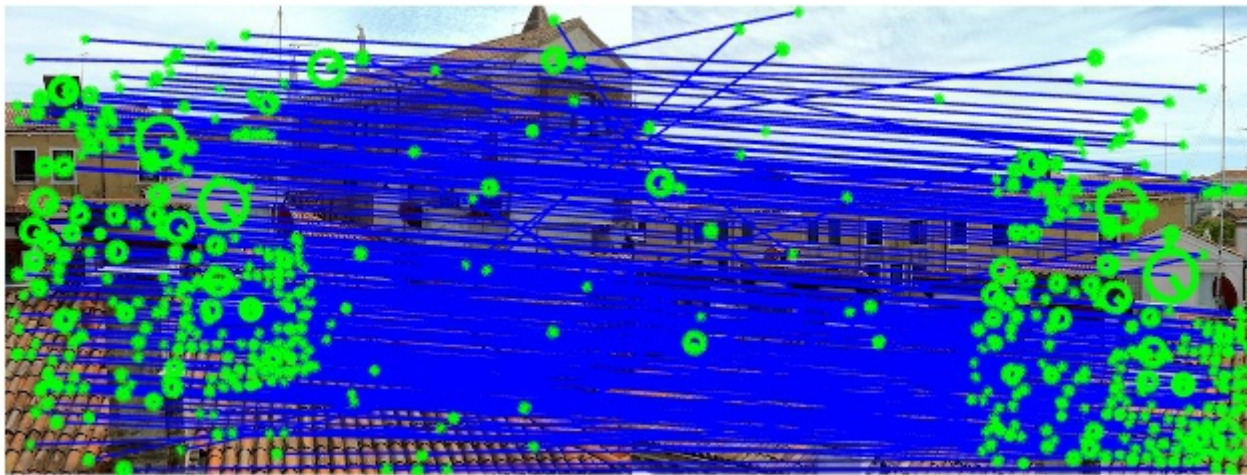
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Source: <http://www.vlfeat.org/overview/sift.html>

Feature Descriptors

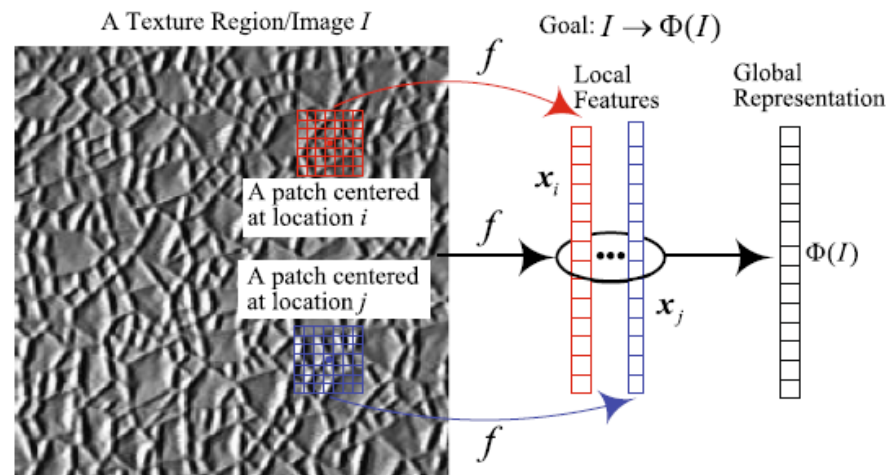
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Feature Descriptors

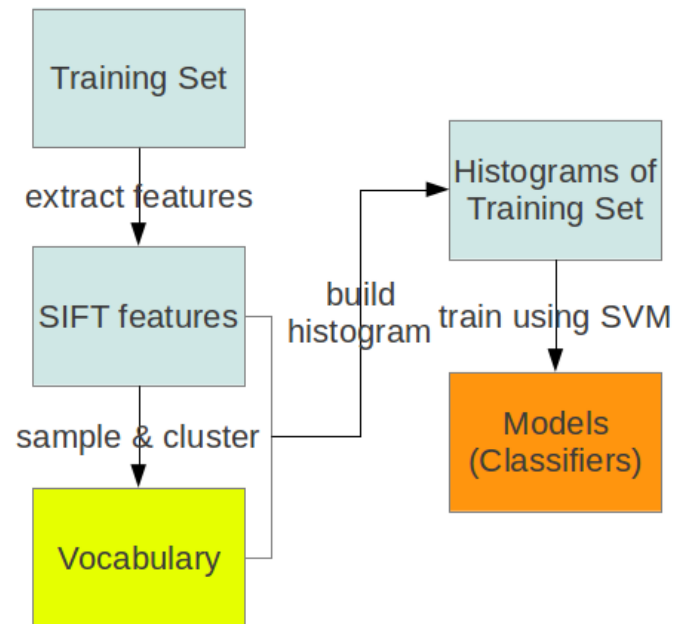
- Scale-invariant feature transform (SIFT):
 - How to use SIFT for texture classification, with so many keypoints?
 - Answer: Feature Encoding
 - Integrate the local features (i.e. SIFT keypoint descriptors) of an image into a global vector to represent the whole image



Source: L. Liu et al. *From BoW to CNN: two decades of texture representation for texture classification*. International Journal of Computer Vision, 2019.

Feature Encoding

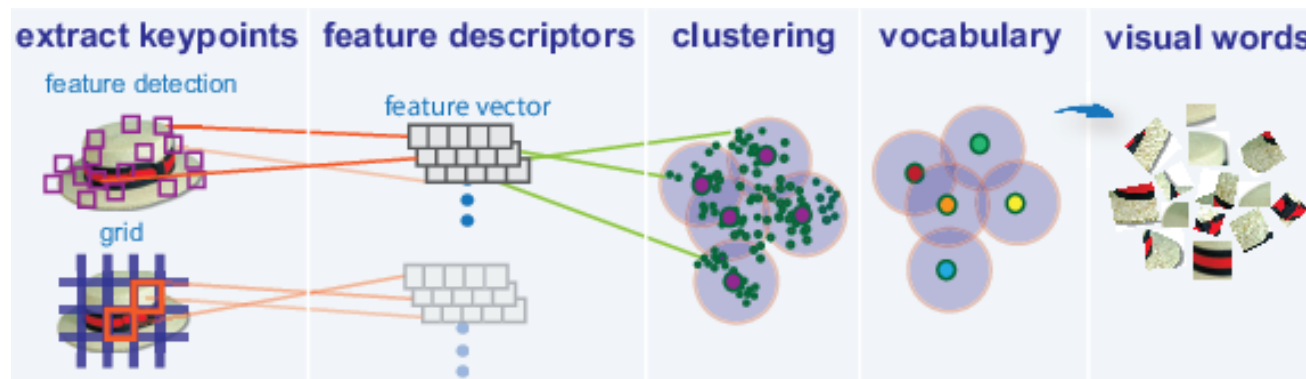
- The most popular method: Bag-of-Words (BoW)
- Local image features are encoded into a histogram to represent the overall image characteristics.



Source: <http://cs.brown.edu/courses/cs143/2011/results/proj3/hangsu/>

Feature Encoding

- Bag-of-Words (BoW)
 - Step 1: create the vocabulary from the set of local descriptors (e.g. SIFT keypoint descriptors) extracted from the training data
 - This vocabulary represents the categories of local descriptors.



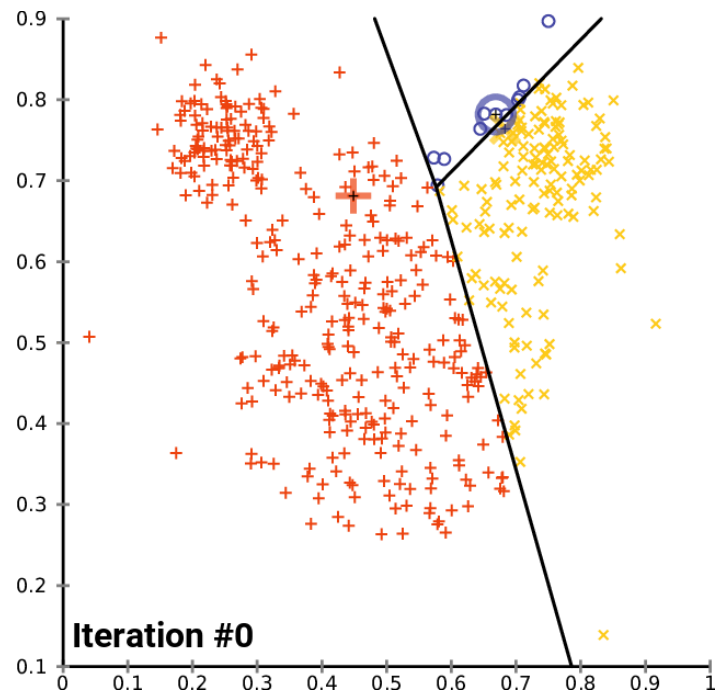
Source: <https://au.mathworks.com/help/vision/ug/image-classification-with-bag-of-visual-words.html>

Feature Encoding

- Bag-of-Words (BoW)
 - Main technique used to create the vocabulary: *k-means clustering*
 - K-means clustering is one of the simplest and most popular unsupervised learning approaches that perform automatic clustering (i.e. partitioning) of the training data into multiple categories

Feature Encoding

- Bag-of-Words (BoW)
 - K-means clustering: the data are iteratively grouped into different clusters based on their distances from the cluster centres.



Source: https://en.wikipedia.org/wiki/K-means_clustering

Feature Encoding

- Bag-of-Words (BoW)
 - The cluster centres become the “vocabularies”, i.e. a set of “visual words”.
 - Individual local feature descriptor (e.g. SIFT keypoint descriptor) is assigned to one visual word with the smallest distance.



Source: <https://au.mathworks.com/help/vision/ug/image-classification-with-bag-of-visual-words.html>

Source: https://en.wikipedia.org/wiki/K-means_clustering

Feature Encoding

- Bag-of-Words (BoW)
 - For an image, the number of local descriptors assigned to each visual word is computed.
 - The numbers are concatenated into a vector => the BoW representation of the image



Source: <https://au.mathworks.com/help/vision/ug/image-classification-with-bag-of-visual-words.html>

Source: https://en.wikipedia.org/wiki/K-means_clustering

Feature Encoding

- Bag-of-Words (BoW)
 - BoW is often considered as a histogram representation since it counts the occurrence frequencies of visual words.
 - The BoW feature vectors are derived for all training images, and used to train a classifier to perform the texture classification.



Source: <https://au.mathworks.com/help/vision/ug/image-classification-with-bag-of-visual-words.html>

Source: https://en.wikipedia.org/wiki/K-means_clustering

Feature Encoding

- BoW recap:
 - Extract a set of local feature descriptors
 - K-means clustering and generate visual words
 - Assign local descriptors to one nearest visual word
 - Count the number of occurrences of the visual words and obtain the histogram feature representation
- What are the main limitations with BoW?

Feature Encoding

- Go beyond BoW
 - The very powerful model: *Fisher Vector (FV) Encoding*
- Main differences from BoW:
 - Use Gaussian Mixture Models instead of k-means clustering
 - Soft assignment instead of assigning to the nearest visual word
 - Computation of first and second-order distances
 - Resulting in a very high-dimensional feature representation (e.g. 64K elements in the final feature vector)

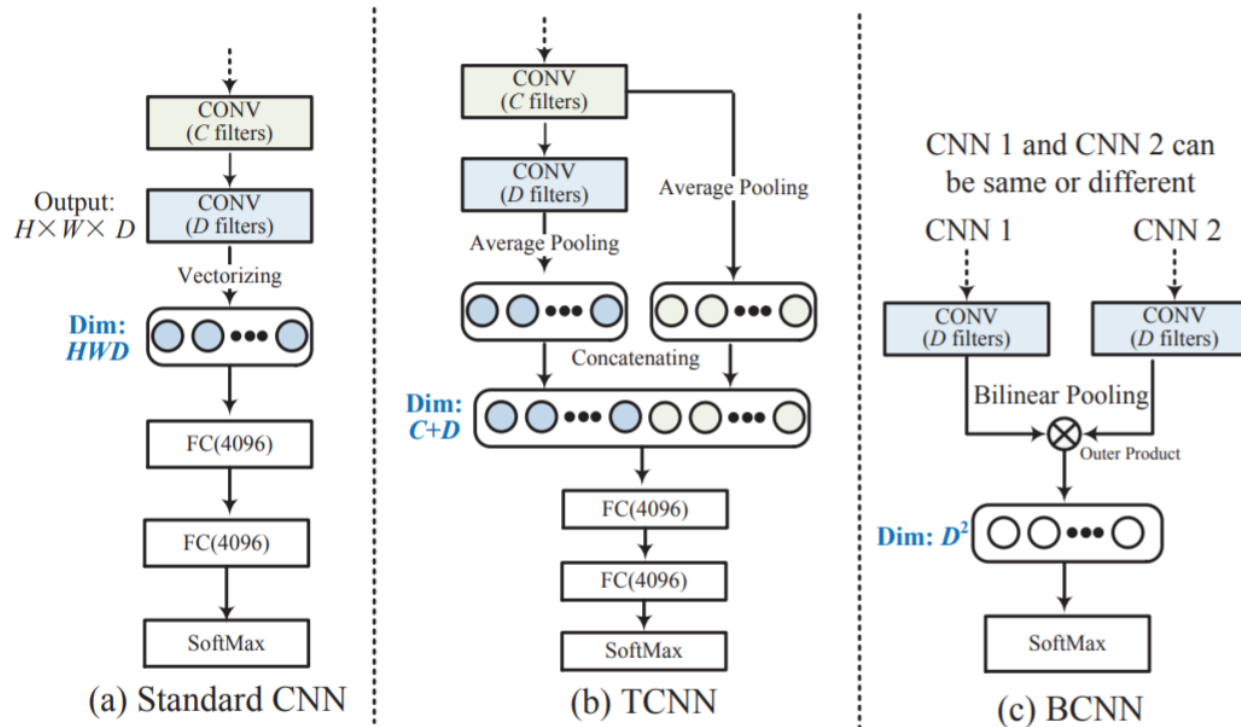
Feature Encoding

- Fisher Vector Encoding:
 - Works best with the Support Vector Machine (SVM) classifier, due to the high-dimensionality of Fisher Vectors.
 - Achieved the highest texture classification performance prior to deep learning approaches
 - Further reading:
 - M. Cimpoi et al. *Describing textures in the wild*. CVPR, 2014.



Deep Learning

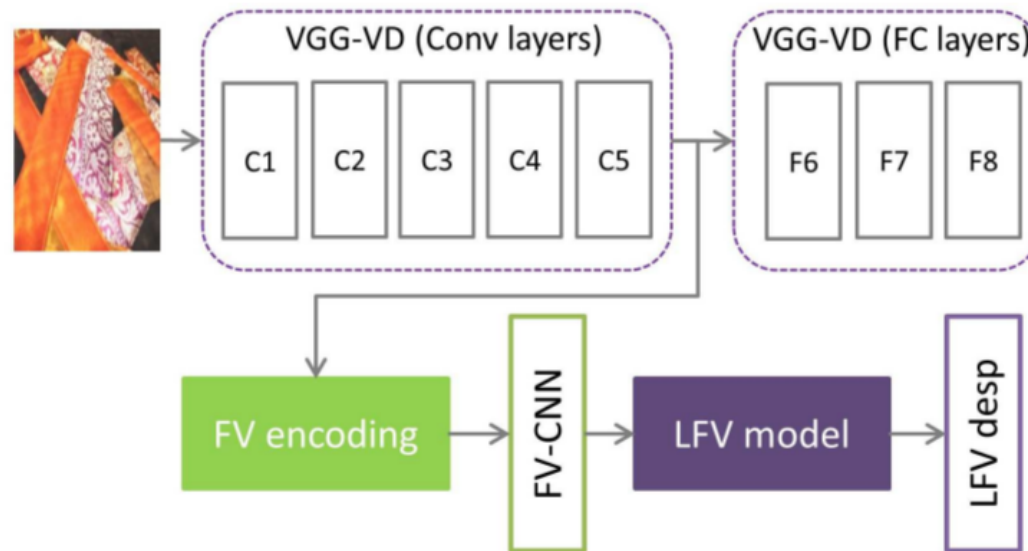
- Standard CNN models have been customised for better texture classification results.



Source: L. Liu et al. *From BoW to CNN: two decades of texture representation for texture classification*. International Journal of Computer Vision, 2019.

Deep Learning

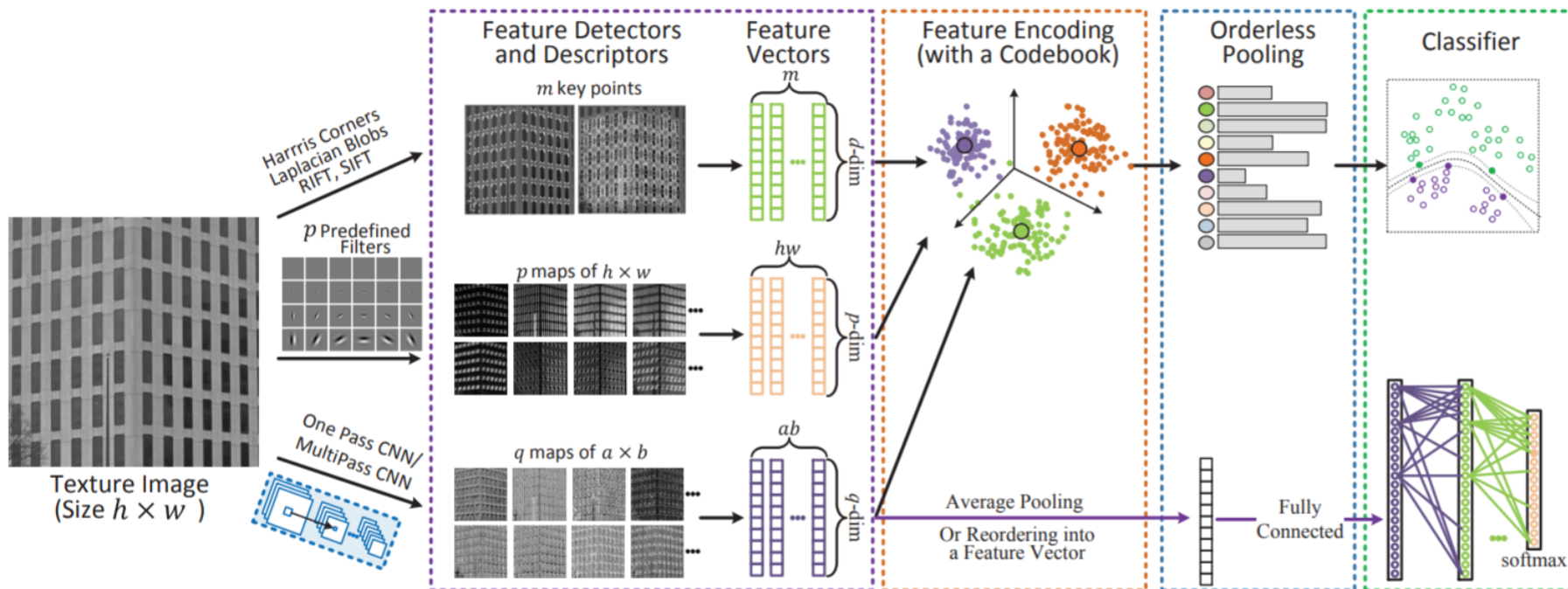
- CNN models have been integrated with Fisher Vector encoding to provide state-of-the-art texture classification performance.



Source: Y. Song et al. *Locally-transferred Fisher Vectors for texture classification*. ICCV, 2017.

Feature Encoding vs. Deep Learning

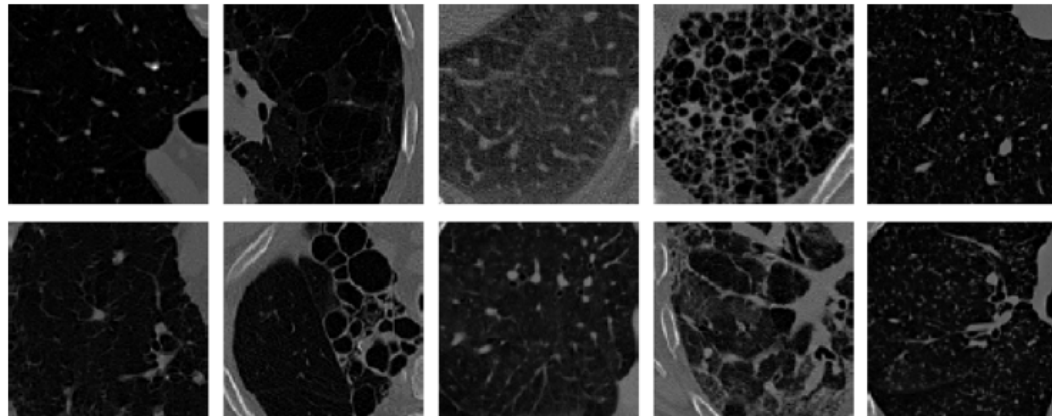
- A unified model:



Source: L. Liu et al. *From BoW to CNN: two decades of texture representation for texture classification*. International Journal of Computer Vision, 2019.

Application

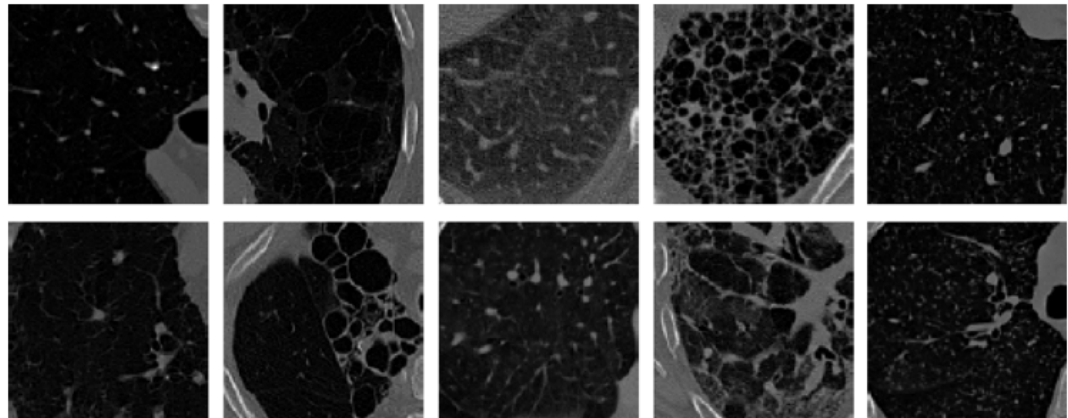
- Classification of interstitial lung disease (ILD):
 - ILD represents a group of more than 150 disorders of the lung parenchyma.
 - Highly challenging with high intra-class variation and inter-class ambiguity



Normal Emphysema Ground glass Fibrosis Micronodules

Application

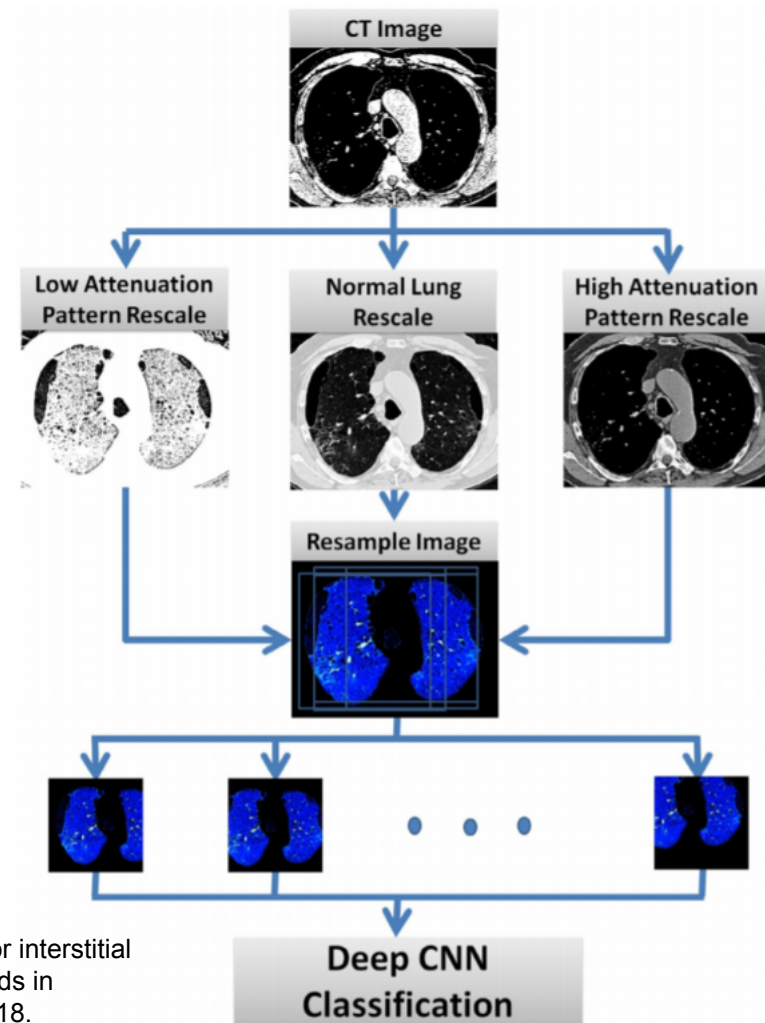
- Classification of ILD:
 - Many approaches have been developed over the past a few years.
 - Mainly two categories of approaches:
 - Texture feature encoding + classification
 - Deep learning



Normal Emphysema Ground glass Fibrosis Micronodules

Application

- Classification of ILD:
 - Integrating domain knowledge with deep learning has shown highly promising results on ILD classification
 - This is an important trend in AI and deep learning research.



Source: M. Gao et al. Holistic classification of CT attenuation patterns for interstitial lung diseases via deep convolutional neural networks. Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization, 2018.

Summary

- Texture classification is an important and fundamental problem in many applications.
- The methods have evolved from morphological representation to deep learning.
- Classification accuracy is still unsatisfactory in many cases.
- Major design choices to make:
 - Pre-processing?
 - Feature representation?
 - Classifier?
 - Training data?
 - Evaluation metrics?

Further Reading

- L. Liu et al. *From BoW to CNN: two decades of texture representation for texture classification*. International Journal of Computer Vision, 2019.
- A very useful computer vision toolkit with concise tutorials.
<http://www.vlfeat.org/>
- A list of popular classifier models.
<https://www.dezyre.com/article/top-10-machine-learning-algorithms/202>

Further Reading

- K. Simonyan et al. *Fisher vector faces in the wild*. BMVC, 2013.

